
Bayesian Inference

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Fundamentals of Bayesian Inference

Consider an observed dataset $\mathcal{D}$ and a statistical model parameterized by an unknown parameter vector $\boldsymbol{\theta} \in \Theta$, where Θ denotes the parameter space. In the Bayesian framework, uncertainty about $\boldsymbol{\theta}$ before observing the data is represented by a prior distribution $p(\boldsymbol{\theta})$.

Given a particular value of $\boldsymbol{\theta}$, the observed data are characterized by the conditional distribution

$$p(\mathcal{D} \mid \boldsymbol{\theta}),$$

which, when viewed as a function of $\boldsymbol{\theta}$ for fixed $\mathcal{D}$, defines the likelihood function.

The joint distribution of $\boldsymbol{\theta}$ and $\mathcal{D}$ can be factorized in two equivalent ways:

$$p(\boldsymbol{\theta}, \mathcal{D}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta}), \quad p(\boldsymbol{\theta}, \mathcal{D}) = p(\boldsymbol{\theta} \mid \mathcal{D})p(\mathcal{D}).$$

Equating the two factorizations yields

$$p(\boldsymbol{\theta} \mid \mathcal{D})p(\mathcal{D}) = p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta}),$$

and therefore Bayes' theorem gives

$$\boxed{p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}}. \quad (1.1)$$

The denominator is obtained by marginalizing the joint distribution over the parameter space:

$$\boxed{p(\mathcal{D}) = \int_{\Theta} p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}}. \quad (1.2)$$

The quantity $p(\mathcal{D})$ is referred to as the marginal likelihood, or model evidence. It normalizes the posterior distribution so that

$$\int_{\Theta} p(\boldsymbol{\theta} \mid \mathcal{D})d\boldsymbol{\theta} = 1.$$

For a fixed observed dataset $\mathcal{D}$, the marginal likelihood $p(\mathcal{D})$ does not depend on $\boldsymbol{\theta}$. Hence, the posterior



distribution can be written up to a proportionality constant as

$$\boxed{p(\boldsymbol{\theta} \mid \mathcal{D}) \propto p(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}. \quad (1.3)$$

Thus, the basic principle of Bayesian inference can be summarized as

$$\boxed{\text{Posterior} \propto \text{Likelihood} \times \text{Prior}}.$$

Bayesian inference can also be used to quantify uncertainty about future observations. Let $\mathcal{D}^*$ denote a future observation or a set of future observations. By marginalizing over the uncertainty in $\boldsymbol{\theta}$, the posterior predictive distribution is

$$p(\mathcal{D}^* \mid \mathcal{D}) = \int_{\Theta} p(\mathcal{D}^* \mid \boldsymbol{\theta}, \mathcal{D})p(\boldsymbol{\theta} \mid \mathcal{D})d\boldsymbol{\theta}.$$

If $\mathcal{D}^*$ is conditionally independent of $\mathcal{D}$ given $\boldsymbol{\theta}$, i.e., $\mathcal{D}^* \perp \mathcal{D} \mid \boldsymbol{\theta}$, then

$$p(\mathcal{D}^* \mid \boldsymbol{\theta}, \mathcal{D}) = p(\mathcal{D}^* \mid \boldsymbol{\theta}),$$

and the posterior predictive distribution reduces to

$$\boxed{p(\mathcal{D}^* \mid \mathcal{D}) = \int_{\Theta} p(\mathcal{D}^* \mid \boldsymbol{\theta})p(\boldsymbol{\theta} \mid \mathcal{D})d\boldsymbol{\theta}}. \quad (1.4)$$

The posterior predictive distribution therefore propagates uncertainty in the parameters into uncertainty about future observations by averaging the predictive distribution over the posterior distribution of $\boldsymbol{\theta}$.

Bayesian Regression

§ 2.1

General Framework of Bayesian Regression

Consider a training dataset

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N,$$

consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional predictor vector associated with the i -th observation, and $y_i \in \mathbb{R}$ denotes the corresponding scalar response. Define the design matrix $\mathbf{X}$, response vector $\mathbf{y}$, and regression coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

Given the predictor vector $\mathbf{x}_i$ and the regression coefficient vector $\mathbf{w}$, the response y_i is characterized by the conditional distribution

$$p(y_i \mid \mathbf{x}_i, \mathbf{w}).$$

Assuming that the observations are conditionally independent given $\mathbf{w}$, the likelihood of the observed response vector is

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \mathbf{w}).$$

In the Bayesian framework, the regression coefficient vector $\mathbf{w}$ is treated as a random variable and assigned a prior distribution $p(\mathbf{w})$. Given the observed dataset $\mathcal{D} = (\mathbf{X}, \mathbf{y})$, Bayes' theorem gives

$$p(\mathbf{w} \mid \mathcal{D}) = p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})}.$$

Using

$$p(\mathbf{X}, \mathbf{y}) = p(\mathbf{y} \mid \mathbf{X})p(\mathbf{X}),$$

the posterior distribution can be written as

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{y} \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})} = \frac{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w} \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})}.$$

Assuming that the prior distribution of $\mathbf{w}$ is independent of the design matrix $\mathbf{X}$, i.e.,

$$p(\mathbf{w} \mid \mathbf{X}) = p(\mathbf{w}),$$

the posterior distribution becomes

$$\boxed{p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}}. \quad (2.1)$$

The denominator

$$\boxed{p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})d\mathbf{w}} \quad (2.2)$$

is referred to as the marginal likelihood, or model evidence.

Since it does not depend on $\mathbf{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$\boxed{p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})}, \quad (2.3)$$

where $p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})$ represents the likelihood and $p(\mathbf{w})$ represents the prior distribution. Therefore, the fundamental principle of Bayesian inference can be summarized as

$$\boxed{\text{Posterior} \propto \text{Likelihood} \times \text{Prior}}.$$

The objective of Bayesian regression is to infer the posterior distribution of the regression coefficient vector $\mathbf{w}$ given the observed dataset $\mathcal{D}$, and subsequently to obtain the posterior predictive distribution of the response y^* for a new observation with predictor vector $\mathbf{x}^*$.

$$p(\mathbf{y}^* \mid \mathbf{X}^*, \mathcal{D}) = \int p(\mathbf{y}^* \mid \mathbf{X}^*, \mathbf{w})p(\mathbf{w} \mid \mathcal{D})d\mathbf{w}.$$

§ 2.2

Bayesian Linear Regression

2.2.1 Basic Settings

Consider a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$



consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional feature vector associated with the i -th observation and $y_i \in \mathbb{R}$ denotes the corresponding scalar response.

Define design matrix $\mathbf{X}$, target vector $\mathbf{y}$, and coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

Under the linear regression model, each observation satisfies

$$y_i = \mathbf{w}^\top \mathbf{x}_i + \epsilon_i,$$

where the noise terms are assumed to be independent and identically distributed as

$$\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, N.$$

Define the noise vector

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_N \end{bmatrix}.$$

It follows that

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix.

For the entire dataset, the linear regression model can therefore be written as

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}.$$

Consequently,

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) \sim \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N).$$

2.2.2 Inference

Bayesian method includes two basic steps: inference and prediction, where inference is to estimate posterior $\mathbf{w}$, while prediction is to predict y^* given $\mathbf{x}^*$.

Model Establishment

In Bayesian regression, the coefficient vector $\mathbf{w}$ is treated as a random variable and assigned a prior distribution $p(\mathbf{w})$. We assume that the prior distribution of $\mathbf{w}$ does not depend on $\mathbf{X}$, i.e., $w \perp X$, and thus

$$p(\mathbf{w} \mid \mathbf{X}) = p(\mathbf{w}).$$

Given the training dataset, Bayesian formula is decomposed as:

$$p(\mathbf{w} \mid \mathcal{D}) = p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})} = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{y} \mid \mathbf{X})p(\mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{y} \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})} = \frac{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w} \mid \mathbf{X})}{p(\mathbf{y} \mid \mathbf{X})} = \frac{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})},$$

where the fifth equality follows from

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w}, \mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w} \mid \mathbf{X})p(\mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{y} \mid \mathbf{X})}{p(\mathbf{w} \mid \mathbf{X})},$$

and the last equality follows from independency of prior coefficient and design matrix.

The denominator

$$p(\mathbf{y} \mid \mathbf{X}) = \int p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w})d\mathbf{w}$$

is the marginal likelihood. Since it does not depend on $\mathbf{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})p(\mathbf{w}),$$

where $p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})$ is a likelihood function and $p(\mathbf{w})$ is a prior function.

Suppose that a zero-mean Gaussian prior is imposed on the coefficient vector:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \Sigma_p),$$

where $\Sigma_p \in \mathbb{R}^{p \times p}$ denotes the prior covariance matrix.

Since both the likelihood and the prior are Gaussian, their conjugacy implies that the posterior distribution of $\mathbf{w}$ is also Gaussian:

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_w, \Sigma_w)$$

Thus,

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \Sigma_p).$$

The remaining task is therefore to derive the posterior mean $\boldsymbol{\mu}_w$ and posterior covariance Σ_w .

Model Solution

To derive the posterior distribution, we first write the product of the likelihood and the prior explicitly:

$$\mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \Sigma_p) = \frac{1}{(2\pi)^{N/2} \sigma^N} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w})\right) \frac{1}{(2\pi)^{p/2} |\Sigma_p|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{w})^\top \Sigma_p^{-1} \mathbf{w}\right).$$



Thus,

$$\begin{aligned}
p(\mathbf{w}|\mathbf{X}, \mathbf{y}) &\propto \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top(\mathbf{y} - \mathbf{X}\mathbf{w}) - \frac{1}{2}(\mathbf{w})^\top \Sigma_p^{-1} \mathbf{w}\right) \\
&= \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \mathbf{w}) - \frac{1}{2}(\mathbf{w})^\top \Sigma_p^{-1} \mathbf{w}\right) \\
&= \exp\left(-\frac{1}{2\sigma^2} \mathbf{y}^\top \mathbf{y} + \frac{1}{\sigma^2} \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} - \frac{1}{2} \mathbf{w}^\top \left(\frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \Sigma_p^{-1}\right) \mathbf{w}\right).
\end{aligned}$$

Comparing this expression with the kernel of a multivariate Gaussian distribution,

$$\exp\left(-\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu}_w)^\top \Sigma_w^{-1}(\mathbf{w} - \boldsymbol{\mu}_w)\right) = \exp\left(-\frac{1}{2} \mathbf{w}^\top \Sigma_w^{-1} \mathbf{w} + \mathbf{w}^\top \Sigma_w^{-1} \boldsymbol{\mu}_w - \frac{1}{2} \boldsymbol{\mu}_w^\top \Sigma_w^{-1} \boldsymbol{\mu}_w\right),$$

we obtain

$$\Sigma_w^{-1} = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \Sigma_p^{-1} = \mathbf{A}.$$

Matching the linear terms gives

$$\mathbf{w}^\top \Sigma_w^{-1} \boldsymbol{\mu}_w = \frac{1}{\sigma^2} \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} \Leftrightarrow \mathbf{A} \boldsymbol{\mu}_w = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{y},$$

and therefore

$$\boldsymbol{\mu}_w = \frac{1}{\sigma^2} \mathbf{A}^{-1} \mathbf{X}^\top \mathbf{y}.$$

Hence, the posterior distribution has the closed-form solution

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \sim \mathcal{N}(\boldsymbol{\mu}_w, \Sigma_w),$$

where

$$\Sigma_w^{-1} = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \Sigma_p^{-1}, \quad \boldsymbol{\mu}_w = \frac{1}{\sigma^2} \Sigma_w \mathbf{X}^\top \mathbf{y}.$$

2.2.3 Prediction

After obtaining the posterior distribution of the coefficient vector, Bayesian linear regression can be used to predict the responses associated with new observations.

Let $\mathbf{X}^* \in \mathbb{R}^{N^* \times p}$ denote the design matrix for N^* new observations, and let $\mathbf{y}^* \in \mathbb{R}^{N^*}$ denote the corresponding target vector to be predicted.

Given the posterior distribution, the posterior predictive distribution is obtained by marginalizing over the uncertainty in $\mathbf{w}$:

$$p(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \int p(\mathbf{y}^* | \mathbf{X}^*, \mathbf{w}) p(\mathbf{w} | \mathcal{D}) d\mathbf{w}.$$

Thus, we have

$$p(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \mathcal{N}(\mathbf{X}^* \boldsymbol{\mu}_w, \mathbf{X}^* \Sigma_w (\mathbf{X}^*)^\top + \sigma^2 \mathbf{I}_{N^*}).$$



Therefore, the predictive mean and covariance are

$$\mathbb{E}[\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}] = \mathbf{X}^* \boldsymbol{\mu}_w,$$

and

$$\text{Cov}(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \underbrace{\mathbf{X}^* \boldsymbol{\Sigma}_w (\mathbf{X}^*)^\top}_{\text{parameter uncertainty}} + \underbrace{\sigma^2 \mathbf{I}_{N^*}}_{\text{observation noise}}.$$

§ 2.3

Bayesian Logistic Regression

2.3.1 Basic Settings

Consider a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional feature vector associated with the i -th observation and $y_i \in \mathbb{B} = \{0, 1\}$ denotes the corresponding binary classification, like yes/no and success/failure.

Define design matrix $\mathbf{X}$, target vector $\mathbf{y}$, and coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

The objective of logistic regression is to model the conditional probability $p(y_i | \mathbf{x}_i, \mathbf{w})$ rather than directly modeling the binary response y_i .

Since y_i is binary, we assume that

$$y_i | \mathbf{x}_i, \mathbf{w} \sim \text{Bernoulli}(p_i),$$

where

$$p_i = p(y_i = 1 | \mathbf{x}_i, \mathbf{w}), \quad 1 - p_i = p(y_i = 0 | \mathbf{x}_i, \mathbf{w}).$$

Recall that in linear regression, the response is modeled as

$$y_i = \mathbf{w}^\top \mathbf{x}_i + \epsilon_i,$$

where the linear predictor $\mathbf{w}^\top \mathbf{x}_i$ can take any real value $\mathbf{w}^\top \mathbf{x}_i \in (-\infty, +\infty)$. In contrast, a probability



must satisfy $p_i \in (0, 1)$. Therefore, we need a function that maps the linear predictor from $\mathbb{R}$ to the interval $(0, 1)$.

In logistic regression, this mapping is provided by the logistic function, also known as the sigmoid function:

$$p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x}_i)} \in (0, 1).$$

Equivalently, the probability of observing $y_i \in 0, 1$ can be written as

$$p(y_i | \mathbf{x}_i, \mathbf{w}) = p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Assuming that the observations are conditionally independent given $\mathbf{w}$ and $\mathbf{X}$, the likelihood of the entire dataset is

$$p(\mathbf{y} | \mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i} = \prod_{i=1}^N \sigma(\mathbf{w}^\top \mathbf{x}_i)^{y_i} (1 - \sigma(\mathbf{w}^\top \mathbf{x}_i))^{1-y_i}.$$

2.3.2 Inference and Prediction

In Bayesian regression, the coefficient vector $\mathbf{w}$ is treated as a random variable and assigned a prior distribution $p(\mathbf{w})$. We assume that the prior distribution of $\mathbf{w}$ does not depend on $\mathbf{X}$, and thus $p(\mathbf{w} | \mathbf{X}) = p(\mathbf{w})$.

Given the training dataset, Bayesian formula is decomposed as:

$$p(\mathbf{w} | \mathcal{D}) = p(\mathbf{w} | \mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})} = \frac{p(\mathbf{w}, \mathbf{y} | \mathbf{X})}{p(\mathbf{y} | \mathbf{X})} = \frac{p(\mathbf{y} | \mathbf{X}, \mathbf{w}) p(\mathbf{w} | \mathbf{X})}{p(\mathbf{y} | \mathbf{X})} = \frac{p(\mathbf{y} | \mathbf{X}, \mathbf{w}) p(\mathbf{w})}{p(\mathbf{y} | \mathbf{X})},$$

where the fifth equality follows from

$$p(\mathbf{y} | \mathbf{X}, \mathbf{w}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w}, \mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w} | \mathbf{X}) p(\mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{y} | \mathbf{X})}{p(\mathbf{w} | \mathbf{X})},$$

and the last equality follows from independency of prior coefficient and design matrix.

The denominator

$$p(\mathbf{y} | \mathbf{X}) = \int p(\mathbf{y} | \mathbf{X}, \mathbf{w}) p(\mathbf{w}) d\mathbf{w}$$

is the marginal likelihood. Since it does not depend on $\mathbf{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$p(\mathbf{w} | \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} | \mathbf{X}, \mathbf{w}) p(\mathbf{w}),$$

where $p(\mathbf{y} | \mathbf{X}, \mathbf{w})$ is a likelihood function and $p(\mathbf{w})$ is a prior function.

In general, we do not impose a particular parametric form on the prior distribution at this stage. Although the posterior is fully specified by the likelihood and the prior, it does not necessarily admit a closed-form

expression. In particular, evaluating the normalizing constant

$$p(\mathbf{y}|\mathbf{X}) = \int p(\mathbf{y} | \mathbf{X}, \mathbf{w})p(\mathbf{w})d\mathbf{w}$$

may be analytically intractable. Therefore, rather than evaluating the posterior density explicitly, Bayesian inference is often performed using numerical approximation methods.

A common approach is to generate a collection of samples

$$\mathbf{w}^{(1)}, \mathbf{w}^{(2)}, \dots, \mathbf{w}^{(S)} \sim p(\mathbf{w}|\mathbf{X}, \mathbf{y}),$$

or approximately from this posterior distribution, using sampling-based inference methods such as Markov chain Monte Carlo (MCMC). These samples provide an empirical approximation to the posterior distribution and can be used to estimate posterior quantities of interest. For example, for an arbitrary function $g(\mathbf{w})$,

$$\mathbb{E}[g(\mathbf{w}) | \mathbf{X}, \mathbf{y}] = \int g(\mathbf{w})p(\mathbf{w} | \mathbf{X}, \mathbf{y})d\mathbf{w} \approx \frac{1}{S} \sum_{s=1}^S g(\mathbf{w}^{(s)}).$$

For a new observation with covariate vector $\mathbf{X}^*$, prediction is based on the posterior predictive distribution

$$p(\mathbf{y}^*|\mathbf{X}^*, \mathcal{D}) = \int p(\mathbf{y}^*|\mathbf{X}^*, \mathbf{w})p(\mathbf{w}|\mathcal{D})d\mathbf{w}.$$

Using posterior samples, this integral can be approximated by Monte Carlo integration as

$$p(\mathbf{y}^*|\mathbf{X}^*, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^S p(\mathbf{y}^*|\mathbf{X}^*, \mathbf{w}^{(s)})$$

Bayesian Filter

§ 3.1

General Framework

3.1.1 Basic Settings

Let

$$\mathbf{x}_t \in \mathbb{R}^{n_x}$$

denote the state vector at time t , which represents the unknown quantity we want to infer.

At each time t , we have observation

$$\mathbf{z}_t \in \mathbb{R}^{n_z},$$

which contains information about the hidden state $\mathbf{x}_t$, but may be noisy and uncertain. For convenience, define the observation history up to time t as

$$\mathbf{z}_{1:t} = \{\mathbf{z}_1, \dots, \mathbf{z}_t\}$$

Similarly, suppose that a control input is available at each time t , denoted by

$$\mathbf{u}_t \in \mathbb{R}^{n_u},$$

with the corresponding control history

$$\mathbf{u}_{1:t} = \{\mathbf{u}_1, \dots, \mathbf{u}_t\}.$$

Therefore, the available information evolves sequentially as

$$\mathbf{u}_1, \mathbf{z}_1, \mathbf{u}_2, \mathbf{z}_2, \mathbf{u}_3, \mathbf{z}_3, \dots$$

whereas the states

$$\mathbf{x}_1, \mathbf{x}_2, \dots$$

are not directly observed.

Since subsequent inference depends on the state, the objective is to characterize the uncertainty associated with the random variable $\mathbf{x}_t$ given all available information. This uncertainty is represented by the belief state

$$\text{bel}(\mathbf{x}_t) = p(\mathbf{x}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}).$$

The belief state is therefore a probability distribution over the possible values of $\mathbf{x}_t$ conditioned on the available observation and control histories. It is also commonly referred to as the posterior state distribution, information state, or state of knowledge.

3.1.2 Core Formula of Bayesian Filter

The key idea of the Bayesian filter is to represent uncertainty explicitly using the calculus of probability theory.

At each time step, the hidden state $\mathbf{x}_t$ evolves from the previous state $\mathbf{x}_{t-1}$ under the control input $\mathbf{u}_t$, while the observation $\mathbf{z}_t$ is generated from the corresponding state $\mathbf{x}_t$, as shown in Figure 3.1.

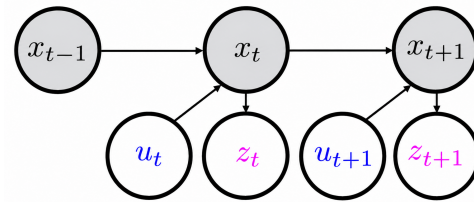


Figure 3.1 Graphical representation of the Bayesian filtering model

However, the belief, $p(\text{current state} | \text{all past information})$, is intractable. This leads to the Markov assumption: future state conditionally independent of past control inputs and observations given present state. We therefore have

$$p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t),$$

$$p(\mathbf{z}_t | \mathbf{x}_t, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{z}_t | \mathbf{x}_t).$$



Using Bayes' rule and the Markov assumptions,

$$\begin{aligned}
\text{bel}(\mathbf{x}_t) &= p(\mathbf{x}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}) = p(\mathbf{x}_t | \mathbf{z}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Definition of belief} \\
&= \frac{p(\mathbf{x}_t, \mathbf{z}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t})}{p(\mathbf{z}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t})} && \leftarrow \text{Condition probability} \\
&= \eta_t p(\mathbf{x}_t, \mathbf{z}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Normalization constant} \\
&= \eta_t p(\mathbf{z}_t | \mathbf{x}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) p(\mathbf{x}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Bayes rule} \\
&= \eta_t p(\mathbf{z}_t | \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Markov assumption} \\
&= \eta_t p(\mathbf{z}_t | \mathbf{x}_t) \int p(\mathbf{x}_t, \mathbf{x}_{t-1} | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) d\mathbf{x}_{t-1} && \leftarrow \text{Marginalization} \\
&= \eta_t p(\mathbf{z}_t | \mathbf{x}_t) \int p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) p(\mathbf{x}_{t-1} | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) d\mathbf{x}_{t-1} && \leftarrow \text{Probability chain rule} \\
&= \eta_t p(\mathbf{z}_t | \mathbf{x}_t) \int p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) p(\mathbf{x}_{t-1} | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) d\mathbf{x}_{t-1} && \leftarrow \text{Unnecessary condition} \\
&= \underbrace{\eta_t p(\mathbf{z}_t | \mathbf{x}_t)}_{\text{correction}} \underbrace{\int p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t) \text{bel}(\mathbf{x}_{t-1}) d\mathbf{x}_{t-1}}_{\text{prediction}}
\end{aligned}$$

where η_t is a normalization constant,

$$\eta_t^{-1} = p(\mathbf{z}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) = \int p(\mathbf{z}_t | \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) d\mathbf{x}_t$$

and $\text{bel}(\mathbf{x}_0) = p(\mathbf{x}_0)$.

The Bayesian filter therefore provides a recursive update approach to calculate belief tractably consisting of two steps: prediction, which propagates the previous belief through the state transition model, and correction, which incorporates the latest observation through the observation model. Algorithm 1 gives the pseudocode of Bayesian filter.

Algorithm 1 Bayesian Filter

Require: Initial belief $\text{bel}(\mathbf{x}_0) = p(\mathbf{x}_0)$, controls $\mathbf{u}_{1:T}$, observations $\mathbf{z}_{1:T}$, transition model $p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t)$, and observation model $p(\mathbf{z}_t | \mathbf{x}_t)$

- 1: **for** $t = 1, 2, \dots, T$ **do**
 - 2: $\overline{\text{bel}}(\mathbf{x}_t) \leftarrow \int p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t) \text{bel}(\mathbf{x}_{t-1}) d\mathbf{x}_{t-1}$ $\triangleright$ Prediction
 - 3: $\eta_t^{-1} \leftarrow \int p(\mathbf{z}_t | \mathbf{x}_t) \overline{\text{bel}}(\mathbf{x}_t) d\mathbf{x}_t$ $\triangleright$ Normalization
 - 4: $\text{bel}(\mathbf{x}_t) \leftarrow \eta_t p(\mathbf{z}_t | \mathbf{x}_t) \overline{\text{bel}}(\mathbf{x}_t)$ $\triangleright$ Correction
 - 5: **end for**
 - 6: **return** $\text{bel}(\mathbf{x}_T)$
-

Kalman Filter

The Kalman filter is a special case of the Bayesian filter in which the state transition and observation models are linear and all uncertainties are Gaussian.

3.2.1 Prediction Step

Suppose that the state transition is linear in the previous state $\mathbf{x}_{t-1}$ and control input $\mathbf{u}_t$:

$$\mathbf{x}_t = \mathbf{F}_t \mathbf{x}_{t-1} + \mathbf{B}_t \mathbf{u}_t + \mathbf{q}_t, \quad \mathbf{q}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \quad t \in \{1, \dots, T\}.$$

Assume that the process noise $\mathbf{q}_t$ is independent of the previous state. Then

$$p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t) = \mathcal{N}(\mathbf{F}_t \mathbf{x}_{t-1} + \mathbf{B}_t \mathbf{u}_t, \mathbf{Q}_t).$$

Suppose that the posterior state distribution at time $t-1$ is Gaussian:

$$p(\mathbf{x}_{t-1} \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = \mathcal{N}(\boldsymbol{\mu}_{t-1|t-1}, \boldsymbol{\Sigma}_{t-1|t-1}),$$

where $\boldsymbol{\mu}_{t-1|t-1}$ and $\boldsymbol{\Sigma}_{t-1|t-1}$ denote the posterior mean and covariance based on all information available up to time $t-1$.

The prediction step of the Bayesian filter is

$$p(\mathbf{x}_t \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) = \int p(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{u}_t) p(\mathbf{x}_{t-1} \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) d\mathbf{x}_{t-1}.$$

Since $\mathbf{x}_{t-1}$ and the independent process noise $\mathbf{q}_t$ are Gaussian, and $\mathbf{x}_t$ is an affine transformation of them, the predictive distribution is also Gaussian:

$$p(\mathbf{x}_t \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) = \mathcal{N}(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}).$$

Its mean is

$$\begin{aligned} \boldsymbol{\mu}_{t|t-1} &= \mathbb{E}[\mathbf{x}_t \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}] && \leftarrow \text{Definition of predictive mean} \\ &= \mathbb{E}[\mathbf{F}_t \mathbf{x}_{t-1} + \mathbf{B}_t \mathbf{u}_t + \mathbf{q}_t] && \leftarrow \text{State transition model} \\ &= \mathbf{F}_t \mathbb{E}[\mathbf{x}_{t-1}] + \mathbf{B}_t \mathbf{u}_t + \mathbb{E}[\mathbf{q}_t] && \leftarrow \text{Linearity of expectation} \\ &= \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \mathbf{u}_t && \leftarrow \text{Zero-mean process noise.} \end{aligned}$$



Its covariance is

$$\begin{aligned}
\boldsymbol{\Sigma}_{t|t-1} &= \text{Cov}(\mathbf{x}_t \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) && \leftarrow \text{Definition of predictive covariance} \\
&= \text{Cov}(\mathbf{F}_t \mathbf{x}_{t-1} + \mathbf{B}_t \mathbf{u}_t + \mathbf{q}_t) && \leftarrow \text{State transition model} \\
&= \text{Cov}(\mathbf{F}_t \mathbf{x}_{t-1} + \mathbf{q}_t) && \leftarrow \text{Deterministic control input} \\
&= \text{Cov}(\mathbf{F}_t \mathbf{x}_{t-1}) + \text{Cov}(\mathbf{q}_t) && \leftarrow \text{Independence of state and process noise} \\
&= \mathbf{F}_t \text{Cov}(\mathbf{x}_{t-1}) \mathbf{F}_t^\top + \mathbf{Q}_t && \leftarrow \text{Covariance under linear transformation} \\
&= \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t.
\end{aligned}$$

Therefore, the prediction step of the Kalman filter is

$$\boldsymbol{\mu}_{t|t-1} = \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \mathbf{u}_t$$

and

$$\boldsymbol{\Sigma}_{t|t-1} = \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t.$$

3.2.2 Correction Step

Suppose that the observation model is also linear:

$$\mathbf{z}_t = \mathbf{H}_t \mathbf{x}_t + \mathbf{r}_t, \quad \mathbf{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \quad t \in \{1, \dots, T\},$$

where the observation noise $\mathbf{r}_t$ is independent of the state $\mathbf{x}_t$. Thus,

$$p(\mathbf{z}_t \mid \mathbf{x}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t}) = p(\mathbf{z}_t \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{H}_t \mathbf{x}_t, \mathbf{R}_t).$$

We first show that the predicted state $\mathbf{x}_t$ and the observation $\mathbf{z}_t$ are jointly Gaussian. From the observation model,

$$\begin{bmatrix} \mathbf{x}_t \\ \mathbf{z}_t \end{bmatrix} = \begin{bmatrix} \mathbf{x}_t \\ \mathbf{H}_t \mathbf{x}_t + \mathbf{r}_t \end{bmatrix} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{H}_t & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}_t \\ \mathbf{r}_t \end{bmatrix}.$$

Before observing $\mathbf{z}_t$,

$$\mathbf{x}_t \sim \mathcal{N}(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}),$$

and

$$\mathbf{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t).$$

Since $\mathbf{x}_t$ and $\mathbf{r}_t$ are independent Gaussian random vectors,

$$\begin{bmatrix} \mathbf{x}_t \\ \mathbf{r}_t \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_{t|t-1} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{t|t-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_t \end{bmatrix} \right).$$

Because a linear transformation of a Gaussian random vector remains Gaussian, $\mathbf{x}_t$ and $\mathbf{z}_t$ are jointly Gaussian.

The predicted-state mean is

$$\boldsymbol{\mu}_{\mathbf{x}t} = \mathbb{E}[\mathbf{x}_t] = \boldsymbol{\mu}_{t|t-1}.$$

The predicted observation mean is

$$\boldsymbol{\mu}_{\mathbf{z}t} = \mathbb{E}[\mathbf{z}_t] = \mathbb{E}[\mathbf{H}_t \mathbf{x}_t + \mathbf{r}_t] = \mathbf{H}_t \mathbb{E}[\mathbf{x}_t] + \mathbb{E}[\mathbf{r}_t] = \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}.$$

The predicted-state covariance is

$$\boldsymbol{\Sigma}_{\mathbf{x}t} = \text{Cov}(\mathbf{x}_t) = \boldsymbol{\Sigma}_{t|t-1}.$$

The cross-covariance between the state and the observation is

$$\boldsymbol{\Sigma}_{\mathbf{x}z t} = \text{Cov}(\mathbf{x}_t, \mathbf{z}_t) = \text{Cov}(\mathbf{x}_t, \mathbf{H}_t \mathbf{x}_t + \mathbf{r}_t) = \text{Cov}(\mathbf{x}_t, \mathbf{H}_t \mathbf{x}_t) + \text{Cov}(\mathbf{x}_t, \mathbf{r}_t) = \text{Cov}(\mathbf{x}_t) \mathbf{H}_t^\top = \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top.$$

Similarly,

$$\boldsymbol{\Sigma}_{\mathbf{z}x t} = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}.$$

Finally, the observation covariance is

$$\boldsymbol{\Sigma}_{\mathbf{z}z t} = \text{Cov}(\mathbf{z}_t) = \text{Cov}(\mathbf{H}_t \mathbf{x}_t + \mathbf{r}_t) = \mathbf{H}_t \text{Cov}(\mathbf{x}_t) \mathbf{H}_t^\top + \text{Cov}(\mathbf{r}_t) = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t.$$

Therefore,

$$\begin{bmatrix} \mathbf{x}_t \\ \mathbf{z}_t \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}_{t|t-1} \\ \mathbf{H}_t \boldsymbol{\mu}_{t|t-1} \end{bmatrix}, \begin{bmatrix} \boldsymbol{\Sigma}_{t|t-1} & \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top \\ \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} & \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t \end{bmatrix} \right).$$

The difference between the actual observation and its predicted mean is called the innovation:

$$\boldsymbol{\nu}_t = \mathbf{z}_t - \mathbb{E}[\mathbf{z}_t] = \mathbf{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}.$$

Before observing $\mathbf{z}_t$, the innovation has zero mean:

$$\mathbb{E}[\boldsymbol{\nu}_t] = \mathbb{E}[\mathbf{z}_t - \mathbb{E}[\mathbf{z}_t]] = \mathbf{0}.$$



Its covariance is

$$\text{Cov}(\boldsymbol{\nu}_t) = \text{Cov}(\mathbf{z}_t) = \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t.$$

We define the innovation covariance

$$\mathbf{S}_t = \underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in the predicted observation}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}}.$$

The two terms in $\mathbf{S}_t$ have distinct interpretations:

$$\underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in the predicted observation}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}}.$$

Since $\mathbf{x}_t$ and $\mathbf{z}_t$ are jointly Gaussian, conditioning on the newly observed $\mathbf{z}_t$ gives a Gaussian posterior:

$$p(\mathbf{x}_t \mid \mathbf{z}_{1:t}, \mathbf{u}_{1:t}) = \mathcal{N}(\boldsymbol{\mu}_{t|t}, \boldsymbol{\Sigma}_{t|t}).$$

For jointly Gaussian random vectors, the conditional mean is

$$\begin{aligned} \boldsymbol{\mu}_{t|t} &= \mathbb{E}[\mathbf{x}_t \mid \mathbf{z}_t] \\ &= \mathbb{E}[\mathbf{x}_t] + \text{Cov}(\mathbf{x}_t, \mathbf{z}_t) \text{Cov}(\mathbf{z}_t)^{-1} (\mathbf{z}_t - \mathbb{E}[\mathbf{z}_t]) \quad \leftarrow \text{Conditional Gaussian formula} \\ &= \boldsymbol{\mu}_{t|t-1} + \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1} (\mathbf{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}) \quad \leftarrow \text{Substitution of mean and covariance.} \end{aligned}$$

Define the Kalman gain

$$\mathbf{K}_t = \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}.$$

The posterior mean can therefore be written compactly as

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t \boldsymbol{\nu}_t.$$

Thus, the new observation corrects the predicted mean by an amount proportional to the innovation $\boldsymbol{\nu}_t$, while the Kalman gain $\mathbf{K}_t$ determines how strongly the observation should influence the estimate.

For jointly Gaussian random vectors, the conditional covariance is

$$\begin{aligned} \boldsymbol{\Sigma}_{t|t} &= \text{Cov}(\mathbf{x}_t \mid \mathbf{z}_t) \\ &= \text{Cov}(\mathbf{x}_t) - \text{Cov}(\mathbf{x}_t, \mathbf{z}_t) \text{Cov}(\mathbf{z}_t)^{-1} \text{Cov}(\mathbf{z}_t, \mathbf{x}_t) \quad \leftarrow \text{Conditional Gaussian formula} \\ &= \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1} \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \quad \leftarrow \text{Substitution of covariance terms} \\ &= \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1}. \end{aligned}$$

Equivalently,

$$\Sigma_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \Sigma_{t|t-1}.$$

Therefore, the correction step of the Kalman filter can be summarized as

$$\begin{aligned} \nu_t &= z_t - \mathbf{H}_t \mu_{t|t-1}, & \mathbf{S}_t &= \mathbf{H}_t \Sigma_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t, & \mathbf{K}_t &= \Sigma_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}, \\ \mu_{t|t} &= \mu_{t|t-1} + \mathbf{K}_t \nu_t, & \Sigma_{t|t} &= (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \Sigma_{t|t-1}. \end{aligned}$$

Therefore, the new observation corrects the predicted state according to the discrepancy between the actual and predicted observations, while the magnitude of the correction depends on both the uncertainty of the state prediction and the observation noise. The complete Kalman filtering procedure is summarized in Algorithm 2.

Algorithm 2 Kalman Filter

Require: Initial posterior mean $\mu_{0|0}$, initial posterior covariance $\Sigma_{0|0}$, controls $\mathbf{u}_{1:T}$, observations $\mathbf{z}_{1:T}$, state transition matrices $\mathbf{F}_{1:T}$ and $\mathbf{B}_{1:T}$, observation matrices $\mathbf{H}_{1:T}$, process noise covariances $\mathbf{Q}_{1:T}$, and observation noise covariances $\mathbf{R}_{1:T}$

- 1: **for** $t = 1, 2, \dots, T$ **do**
- 2: $\mu_{t|t-1} \leftarrow \mathbf{F}_t \mu_{t-1|t-1} + \mathbf{B}_t \mathbf{u}_t$ ▷ Predicted mean
- 3: $\Sigma_{t|t-1} \leftarrow \mathbf{F}_t \Sigma_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t$ ▷ Predicted covariance
- 4: $\nu_t \leftarrow z_t - \mathbf{H}_t \mu_{t|t-1}$ ▷ Innovation
- 5: $\mathbf{S}_t \leftarrow \mathbf{H}_t \Sigma_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t$ ▷ Innovation covariance
- 6: $\mathbf{K}_t \leftarrow \Sigma_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1}$ ▷ Kalman gain
- 7: $\mu_{t|t} \leftarrow \mu_{t|t-1} + \mathbf{K}_t \nu_t$ ▷ Posterior mean
- 8: $\Sigma_{t|t} \leftarrow (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \Sigma_{t|t-1}$ ▷ Posterior covariance
- 9: **end for**
- 10: **return** $\{\mu_{t|t}, \Sigma_{t|t}\}_{t=1}^T$ of $\text{bel}(\mathbf{x}_t)$
